A Mostly Harmless Introduction to Claude Code, Codex and Friends

For Empirical Researchers

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Welcome

Thank you for joining this session of **Workshops for Ukraine**.

- Built for empirical researchers — economists, not software engineers
- Practical and hands-on: what to actually do, not how LLMs are built under the hood

About Me

Daniel Sánchez, MA

- Economist — MA in Economics, Simon Fraser University (Canada)
- Statistician, Government of Alberta — nowcasting economic accounts in R/SQL
- Consultant, Corporate Finance & Digital Transformation, Baker Tilly Ecuador
- Co-director, Laboratorio de Investigación en Desarrollo del Ecuador (LIDE)

Working with agentic AI

- Design GenAI workflows (Claude, ChatGPT, Gemini, Copilot) for due diligence, financial analysis, and reporting
- Build autonomous agents with Claude Code and Cowork for document processing and structured extraction
- Connect agents to everyday tools — Excel, Outlook, Teams — via MCP

Agenda

1. Introduction: why agentic AI
2. What's out there: the landscape of tools
3. How to use agentic AI: mechanics and cost models
4. Quick overview: best model for common users
5. Usages (the core, hands-on section)
6. Cybersecurity: do's and don'ts
7. Multi-agent (tentative)

Introduction: Why Agentic AI

The real bottleneck in most empirical projects is not the econometrics — it's the grind around it:

- Cleaning a messy microdata file for the third time this month
- Chasing down why a regression coefficient changed after "nothing changed"
- Reformatting the same table for a fourth journal's style

That grind is exactly the part a chat window cannot touch — and exactly the part an agent can sit inside.

Vanilla AI vs. Agentic AI

Vanilla AI (ChatGPT, Claude.ai, a chat window)

- You paste in code or data, it replies with text
- You copy the suggestion back out, apply it yourself, rerun it yourself
- It never sees whether its own suggestion actually worked

Agentic AI (Claude Code, Codex, and friends)

- It opens your files, makes the edit itself, runs the script itself

The Landscape: What's Out There

- Terminal / CLI agents: Claude Code, Codex, and friends
- Chat-based assistants vs. agentic tools
- IDE-integrated agents
- What researchers are actually adopting right now

Mechanics and Cost Models

- How an agentic session actually works under the hood
- Context windows, tool calls, and iteration loops
- Subscription vs. API/token-based pricing
- Ballpark costs for a typical research workflow

Quick Overview: Best Model for Common Users

- What "best" depends on: task type, budget, workflow
- Rules of thumb for picking a model/tool as a non-specialist
- When the default choice is good enough

Usages

The core, hands-on section.

1. Code Review

- Catching bugs and silent errors in statistical/data code
- Reviewing before merging or before submitting to a coauthor

2. Literature Review

- Search, summarize, and synthesize across papers
- Where agents help — and where they still hallucinate citations

3. Writing Review

- Drafting and editing papers, referee reports, slides
- Keeping the researcher's voice and claims intact

4. Skills and Agents from Modern Economists

- Reusable prompts, skills, and workflows built by other researchers
- Examples worth adapting

Cybersecurity: Do's and Don'ts

Do

- Use vetted, org-approved tools and check for a data processing agreement
- Keep credentials out of prompts — use secret managers/env vars, never paste tokens or passwords into chat
- Scope MCP/data connections to least privilege (read-only where possible)
- Know what data leaves your machine before granting file or network access

Don't

- Don't paste PII, credentials, or restricted admin microdata into a public chat interface
- Don't grant blanket filesystem or internet access without understanding what the agent can reach
- Don't treat "the agent says it's fine" as a compliance answer
- Don't skip your institution's confidentiality and data-handling

Multi-Agent

(Tentative — first thing to cut if running long)

- Running multiple agents in parallel or in a pipeline
- When one agent isn't enough: division of labor, orchestration
- Practical caveats: cost, coordination, verification overhead

Thank You

- Daniel Sánchez Pazmiño
- Workshops for Ukraine — September 3, 2026